SI-LINK™ AC DFDB-5451 NT
Crosslinkable Polyethylene for Moisture Curable Power Cable Insulation

Overview

SI-LINK™ AC Crosslinkable Polyethylene DFDB-5451 NT is a scorch retardant ethylene- silane-copolymer developed for use in power cable and control cable applications up to 1 kV. DFDB-5451 NT may be crosslinked under ambient conditions after it is extruded with the DFDA-5488 NT catalyst masterbatch. If a black product is required for UV protection, the addition of DFDB-5410 BK is recommended.

Because the catalyst and carbon black masterbatches are shipped separately from the DFDB-5451 NT copolymer, the components are very shelf-stable. Crosslinking of the total system only occurs after hot melt mixing of the components and exposure to moisture and temperature.

Specifications

When DFDB-5451 NT is crosslinked with DFDA-5488 NT, and optionally, DFDB-5410 BK, it should meet typical 1 kV or less low voltage specifications such as those in:

- UL: 854
- ICEA: S-66-524
- CSA: RW90
- IEC: 60502-1

Physical	Nominal Value	(English)	Nominal Value	(SI)	Test Method
Density	0.922	g/cm³	0.922	g/cm³	ASTM D792
Melt Mass-Flow Rate (190°C/2.16 kg)	1.5	g/10 min	1.5	g/10 min	ASTM D1238
Mechanical	**Nominal Value**	**(English)**	**Nominal Value**	**(SI)**	**Test Method**
Tensile Strength [1]	2400	psi	16.5	MPa	ASTM D638
Tensile Elongation [1] (Break)	350	%	350	%	ASTM D638
Thermal	**Nominal Value**	**(English)**	**Nominal Value**	**(SI)**	**Test Method**
Hot Creep - 15 min, 20 N/cm² [2] (302°F (150°C))	< 100	%	< 100	%	ICEA T-28-562
Hot Set - 15 min, 0.2 MPa [2] (392°F (200°C))	< 100	%	< 100	%	IEC 60811-2-1
Aging	**Nominal Value**	**(English)**	**Nominal Value**	**(SI)**	**Test Method**
Retention of Tensile Elongation - 7 days [1]					ASTM D638
250°F (121°C)	95	%	95	%	
Retention of Tensile Strength - 7 days [1]					ASTM D638
250°F (121°C)	90	%	90	%	

Additional Information

Storage:

- The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. The recommended storage conditions are dry conditions with temperatures between 50°F and 86°F (10°C and 30°C). When stored under these conditions, the product may be used by the customer for up to one year from the date of sale or two years from the date of manufacture, whichever comes first. It is recommended that the practice of using the product on a first-in / first-out basis be established.

Extrusion	Nominal Value	(English)	Nominal Value	(SI)
Drying Temperature	140 to 160	°F	60 to 71	°C
Drying Time	4.0 to 6.0	hr	4.0 to 6.0	hr
Melt Temperature	300 to 410	°F	149 to 210	°C

® ™ Trademark of The Dow Chemical Company (“Dow”) or an affiliated company of Dow.
Form No. 400-00092773en
Rev: 2015-05-22

Extrusion Notes

DFDB-5451 NT will extrude with excellent surface quality and without extrusion scorch if the accompanying catalyst masterbatch, DFDA-5488 NT and the carbon black masterbatch, DFDB-5410 BK, are kept dry. It is especially recommended that the carbon masterbatch be dried at 140°F-160°F (60°-70°C) for four to six hours using dehumidified air prior to mixing and extrusion. Melt temperatures in the range of 300°F-410°F (150-210°C) have been successfully used.

After extrusion of the appropriate mixture of this product and its catalyst and carbon black masterbatches, crosslinking can be achieved by allowing moisture to diffuse into the product.

Fabricators can use a hot water bath, sauna or ambient conditions to promote curing following cable manufacture. To achieve a hot creep elongation of 100%, the typical times shown below are required (30 mil [0.76 mm] wall on 14 AWG [2.1 mm²] wire).

- 90°C sauna: 15 minutes
- 23°C, 70% rh: 1.5 days

Specific recommendations for your particular equipment and conditions can be determined by contacting your local Dow Wire and Cable sales representative.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

[1] Crosslinked polyethylene properties (88.7% DFDB-5451 NT, 5% DFDA-5488 NT, and 6.3% DFDB-5410 BK)

[2] Measured on 14 AWG (2.1 mm²) wire with a 30 mil (0.76 mm) wall after 15 minutes curing in 90°C water.
Crosslinked polyethylene properties (88.7% DFDB-5451 NT, 5% DFDA-5488 NT, and 6.3% DFDB-5410 BK)

® ™ Trademark of The Dow Chemical Company (“Dow”) or an affiliated company of Dow.
Form No. 400-00092773en
Rev: 2015-05-22

Product Stewardship

The Dow Chemical Company and its subsidiaries (“Dow”) has a fundamental concern for all who make, distribute, and use its products, and for the environment in which we live. This concern is the basis for our Product Stewardship philosophy by which we assess the safety, health, and environmental information on our products and then take appropriate steps to protect employee and public health and our environment. The success of our Product Stewardship program rests with each and every individual involved with Dow products – from the initial concept and research, to manufacture, use, sale, disposal, and recycle of each product.

Customer Notice

Dow strongly encourages its customers to review both their manufacturing processes and their applications of Dow products from the standpoint of human health and environmental quality to ensure that Dow products are not used in ways for which they are not intended or tested. Dow personnel are available to answer your questions and to provide reasonable technical support. Dow product literature, including safety data sheets, should be consulted prior to use of Dow products. Current safety data sheets are available from Dow.

Medical Applications Policy

NOTICE REGARDING MEDICAL APPLICATION RESTRICTIONS: Dow will not knowingly sell or sample any product or service (“Product”) into any commercial or developmental application that is intended for:

a. long-term or permanent contact with internal bodily fluids or tissues. “Long-term” is contact which exceeds 72 continuous hours;
b. use in cardiac prosthetic devices regardless of the length of time involved (“cardiac prosthetic devices” include, but are not limited to, pacemaker leads and devices, artificial hearts, heart valves, intra-aortic balloons and control systems, and ventricular bypass-assisted devices);
c. use as a critical component in medical devices that support or sustain human life; or
d. use specifically by pregnant women or in applications designed specifically to promote or interfere with human reproduction.

Dow requests that customers considering use of Dow products in medical applications notify Dow so that appropriate assessments may be conducted. Dow does not endorse or claim suitability of its products for specific medical applications. It is the responsibility of the medical device or pharmaceutical manufacturer to determine that the Dow product is safe, lawful, and technically suitable for the intended use. **DOW MAKES NO WARRANTIES, EXPRESS OR IMPLIED, CONCERNING THE SUITABILITY OF ANY DOW PRODUCT FOR USE IN MEDICAL APPLICATIONS.**

Disclaimer

NOTICE: No freedom from infringement of any patent owned by Dow or others is to be inferred. Because use conditions and applicable laws may differ from one location to another and may change with time, the Customer is responsible for determining whether products and the information in this document are appropriate for the Customer’s use and for ensuring that the Customer’s workplace and disposal practices are in compliance with applicable laws and other governmental enactments. Dow assumes no obligation or liability for the information in this document. **NO WARRANTIES ARE GIVEN; ALL IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ARE EXPRESSLY EXCLUDED.**

NOTICE: If products are described as “experimental” or “developmental”: (1) product specifications may not be fully determined; (2) analysis of hazards and caution in handling and use are required; (3) there is greater potential for Dow to change specifications and/or discontinue production; and (4) although Dow may from time to time provide samples of such products, Dow is not obligated to supply or otherwise commercialize such products for any use or application whatsoever.

NOTICE: This data is based on information Dow believes to be reliable, as demonstrated in controlled laboratory testing. They are offered in good faith, but without guarantee, as conditions and method of use of Dow products are beyond Dow’s control. Dow recommends that the prospective user determine the suitability of these materials and suggestions before adopting them on a commercial scale.

To the best of our knowledge, the information contained herein is accurate and reliable as of the date of publication, however we do not assume any liability for the accuracy and completeness of such information.

Additional Information

North America		**Europe/Middle East**	+800-3694-6367
U.S. & Canada:	1-800-441-4369		+31-11567-2626
	1-989-832-1426	Italy:	+800-783-825
Mexico:	+1-800-441-4369		
Latin America		**South Africa**	+800-99-5078
Argentina:	+54-11-4319-0100		
Brazil:	+55-11-5188-9000		
Colombia:	+57-1-219-6000	**Asia Pacific**	+800-7776-7776
Mexico:	+52-55-5201-4700		+603-7965-5392

www.dowplastics.com

This document is intended for use within Africa & Middle East, Asia Pacific, Europe, Latin America, North America

Published: 2006-05-15

© 2017 The Dow Chemical Company

® ™ Trademark of The Dow Chemical Company (“Dow”) or an affiliated company of Dow.

Form No. 400-00092773en
Rev: 2015-05-22